Introduction to Column Vectors

Exam code: 0607

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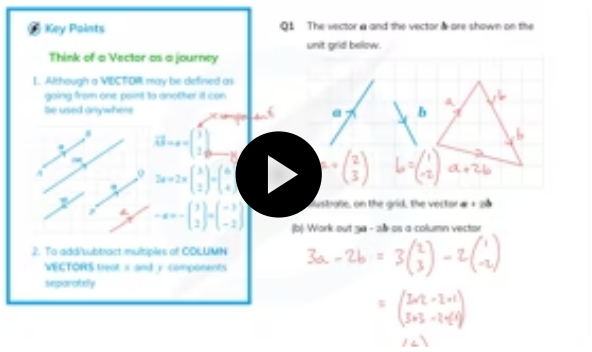

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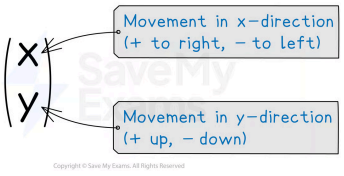
Help me understand

Basic vectors

What are column vectors?

- A **column vector** can be used to **describe** how to get **from one point to another point**
  - This is also called a **translation vector**
  - $\begin{pmatrix} 6 \\ 3 \end{pmatrix}$ means **6 units to the right** and **3 units up**



How do I add and subtract column vectors?

- **Adding and subtracting vectors** is done by looking at the **top** numbers and **bottom** numbers **separately**
- To **add** column vectors
  - Add the top numbers together
  - Add the bottom numbers together

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- Subtract the second top number from the first
- Subtract the second bottom number from the first
  - $\begin{pmatrix} 5 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 5-3 \\ 2-(-1) \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

How do I multiply a vector by a scalar?

- A **scalar** is a **number** not a **vector**
  - It does not have a direction
- To **multiply** a column vector by a **scalar**
  - Multiply the top number by the scalar
  - Multiply the bottom number by the scalar
    - $3\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 3\times 2 \\ 3\times(-1) \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \end{pmatrix}$

How do I write an expression as a single column vector?

- You need to follow the **order of operations**
  - $2\begin{pmatrix} 5 \\ 2 \end{pmatrix} + 5\begin{pmatrix} 3 \\ -1 \end{pmatrix}$
- **STEP 1**
  Multiply each vector by the scalar in front of it
  - $\begin{pmatrix} 2\times 5 \\ 2\times 2 \end{pmatrix} + \begin{pmatrix} 5\times 3 \\ 5\times(-1) \end{pmatrix} = \begin{pmatrix} 10 \\ 4 \end{pmatrix} + \begin{pmatrix} 15 \\ -5 \end{pmatrix}$
- **STEP 2**
  Add or subtract the new column vectors
  - $\begin{pmatrix} 10+15 \\ 4+(-5) \end{pmatrix} = \begin{pmatrix} 25 \\ -1 \end{pmatrix}$

Worked Example

$\mathbf{a} = \begin{pmatrix} p \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$.

Given that $2\mathbf{a} + 3\mathbf{b} = \begin{pmatrix} 4 \\ q \end{pmatrix}$, find the value of p and the value of q.

Answer:

Write the left-side side as one vector
Multiply each vector by the scalar in front of it

$$\begin{pmatrix} 2p \\ 6 \end{pmatrix} + \begin{pmatrix} -6 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ q \end{pmatrix}$$

Add the vectors together

$$\begin{pmatrix} 2p-6 \\ 9 \end{pmatrix} = \begin{pmatrix} 4 \\ q \end{pmatrix}$$

$$2p - 6 = 4$$
$$2p = 10$$
$$p = 5$$

The bottom components are equal

$$9 = q$$

$$p = 5 \text{ and } q = 9$$

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